

1 L2 Spanish Subject Realization: German L1 Learners vs. Spanish L1 Natives

This repository contains the data and R scripts for the study: “The subject realization in L2 Spanish by German L1 speakers: A corpus study”

- **Publication DOI:** <https://doi.org/10.5565/rev/isogloss.437>
 - **Primary Corpus Source:** [CEDEL2](#) (Lozano, 2013)
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1.1 Repository Structure

This project uses the OSF Component strategy. To replicate the results, ensure all files are downloaded into a single working directory.

1.1.1 Component: Data

- **data12s.csv:** Subject realization data for 82 German L1 learners of Spanish across all proficiency levels.
- **datanatives.csv:** A random sample of 17 native Spanish speakers used as a comparative baseline.

1.1.2 Component: Analysis

- **demographics.R:** Script for participant metadata and learner profile summaries.
 - **descriptive.R:** Script for frequency counts.
 - **mupdarfanalysis.R:** The core script implementing the MuPDARF deviation analysis.
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1.2 Data Description

The datasets include sentence-level coding for subject realization in the “**Chaplin**” task (silent film retelling).

1.2.1 Inclusion Criteria

To isolate contexts where subject realization is optional/variable, the following were excluded:

* Imperatives * Indefinite “se” constructions * Impersonal sentences (e.g., hay, llueve)

1.2.2 Key Variables Table

Variable	Description	Values
Subject	Binary realization	NS (Null), RS (Realized)
SubjectType	Detailed classification	dp, nullsub, pron
SubjectPosition	Syntactic position	PRE, POST
Thetarole	Semantic role	agent, pacient
InformationStructure	Discourse status	GI (Given), NE (New)
Proficiency	Learner level	LA to HC

1.3 Reproducibility Instructions

1.3.1 1. Requirements

You will need **R** and the following libraries installed: * partykit * dplyr * ggplot2 * stringr * lattice

1.3.2 2. Setting the Environment

Because this repository does not include an **.Rproj** file, you must manually set your working directory in R to the folder containing the downloaded files:

```
setwd("path/to/your/downloaded/files")
```

1.3.3 3. Analysis Procedure: MuPDARF

The analysis follows the **MuPDARF** (Multifactorial Prediction and Deviation Analysis using Random Forests) method: 1. **Baseline**: A Random Forest is trained on **datanatives.csv** to establish native speaker patterns. 2. **Prediction**: The model predicts the expected native-like choices for the German L1 learners in **data12s.csv**. 3. **Deviation**: Analysis of the statistical deviation between actual learner choices and model-predicted native choices.

1.4 Citation

If you use these scripts or data, please cite the original paper:

Kocher, Anna (2025). The subject realization in L2 Spanish by German L1 speakers: A corpus study. Isogloss. Open Journal of Romance Linguistics, 11(2), 1–34. <https://doi.org/10.5565/rev/isogloss.437> And the corpus:

Lozano, C. (2013). CEDEL2: Design, compilation and reach of a corpus of L2 Spanish. Second Language Research, 29(1), 135-147.